

ALLIANCE KCSE 2025 PREDICTION



POSSIBLE KCSE EXAM



MATHEMATICS

121/2

PAPER 2

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education

INSTRUCTIONS TO CANDIDATES

- a) Write your name, index number and class in the spaces provided above.
- b) The paper contains two sections: **Section I** and **Section II**.
- c) Answer **ALL** the questions in **Section I** and **ANY FIVE** questions from **Section II**.
- d) All working and answers must be written on the question paper in the spaces provided below each question.
- e) Marks may be awarded for correct working even if the answer is wrong.
- f) Negligent and slovenly work will be penalized.
- g) Non-programmable silent electronic calculators and mathematical tables are allowed for use.

For examiners use only

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

Grand

Total

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SECTION I (50 marks) Answer all the questions

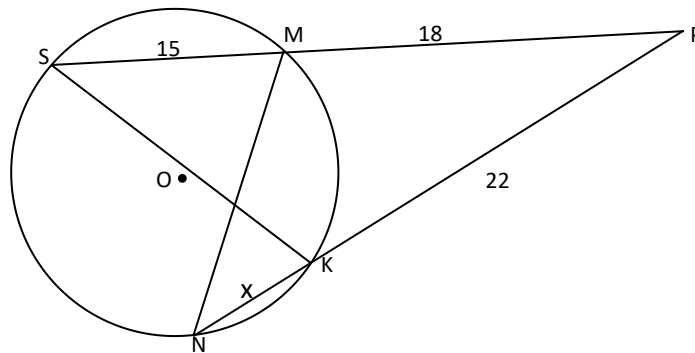
1. Find matrix X given that $AX - X = \begin{pmatrix} -4 \\ -10 \end{pmatrix}$ where $A = \begin{pmatrix} 2 & 3 \\ -2 & 4 \end{pmatrix}$ (3 marks)

2. The cost of providing a commodity consists of transport, labour and raw material in the ratio 8:4:12 respectively. If the transport cost increases by 12 % labour cost 18 % and raw materials by 40 %, find the percentage increase of producing the new commodity (3 marks)

3. Find the number of terms in the series $\frac{8}{81} - \frac{4}{27} + \frac{2}{9} - \dots - 1\frac{11}{16}$ (3marks)

4. Using binomial expansion simplify $\frac{(2+\sqrt{3})^4}{(2-\sqrt{3})^4}$ leaving your answer in the form $a+b\sqrt{c}$ where a, b and c are rational numbers **(3marks)**

5. The figure below shows external intersection of two chords SM and NK which are produced to meet at P. Given that SOK is the diameter of the circle and that SM=15cm, MP=18cm and PK=22cm



- a). Calculate the value of x representing length of chord NK. **(1 marks)**
- b). Calculate the radius of the circle **(2marks)**

6. Determine the equation of a circle if the coordinates of the end points of its diameter are **A(-4,4) and B(6,-12)** leaving your answer in the form of $ax^2 + by^2 + cx + dy + e = 0$ where a, b, c, d and e are integers. **(3 marks)**

7. Make 'c' the subject of the formula if $v^2 = \frac{r}{3} + \sqrt{\frac{1+c^2}{r^2}}$ **(3marks)**

8. Find the area enclosed by the curve $y = 81 - x^2$ and the x -axis using mid-ordinate rule with 9 strips. **(4marks)**

9. Without using a calculator or tables, find the value of x in

(3marks) $\log_{2.5}\left(\frac{1}{x}\right) - 1 - \log_{2.5}(x) = 3\log_{2.5}\left(\frac{1}{x}\right) - \log_{0.5} 2$

10. Solve for ϑ in the equation $6\cos^2 \vartheta - \sin \vartheta - 4 = 0$ in the range $-180^\circ \leq \vartheta \leq 180^\circ$ **(3 marks).**

11. Wambua deposited 6,400 in an investment account which pays 30% interest rate per annum compounded semi – annually for 3 years. Muinde invested one and a half times that of Wambua at $12\frac{1}{2}\%$ per annum simple interest for 6 years. Find whose investment earned more interest and by how much give your answer to the nearest shilling. **(3marks)**

12. (a) Expand and simplify the binomial expression.

(2 Marks)

$$\left(2 - \frac{1}{2}y\right)^5$$

(b) Use the first four terms of the simplified expression in (a) above to evaluate to 5 significant figures.
(1.98)⁵.

(2 Marks)

13. Use logarithms to evaluate

$$\sqrt[3]{\frac{5.27 \times \cos^2 82.48}{\log 6.7}}$$

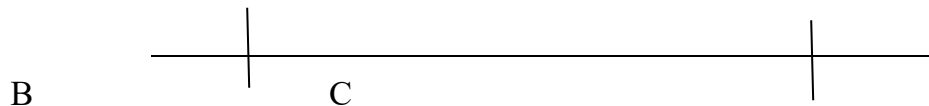
(3marks)

14. Form the quadratic equation whose roots are $x = -\frac{5}{3}$ and $x = 1$

(2 marks)

- 15.** A biased coin is weighted such that its tail is twice as likely to appear as the head. Find the probability that that a tail and a head appear when the coin is tossed twice. **(3marks)**

- 16.** Line BC below is a side of a triangle ABC and also side of a parallelogram BCDE.



Using a ruler and a pair of compasses only construct:

- (i) The triangle ABC given that $\angle ABC = 120^\circ$ and $AB = 6$ cm. **(1 mark)**
- (ii) The parallelogram BCDE whose area is equal to that of the triangle ABC and point E is on line AB. **(3 marks)**

SECTION II (50 marks)

*Answer **only five** questions from this section*

17. ABCD is a quadrilateral with coordinates $A(2, 1)$, $B(3, 2)$, $C(3, 4)$ and $D(0, 3)$. ABCD is mapped onto $A'B'C'D'$ under transformation T given by a shear with x – axis invariant such that $A'(4, 1)$.
- a) Determine the 2×2 transformation matrix representing T and hence determine the coordinates of B' , C' and D' .
(2 marks)

- b) $A'B'C'D'$ is transformed to $A''B''C''D''$ under a transformation H such that $A''(-6, -9)$ and $D''(-12, -15)$. Determine the 2×2 matrix representing H and hence determine the coordinates of B'' and C'' .
(3 marks)

- c) $A''B''C''D''$ mapped onto $A'''B'''C'''D'''$ under a transformation V representing a reflection in the line $y = -x$. Determine the 2×2 matrix representing V and hence determine the coordinates of $A'''B'''C'''D'''$.
(3 marks)

18. The following table shows heights of 200 seedlings of the same species measured to the nearest cm.

Height(cm)	frequency					
70- 79	7					
80-84	30					
85-89	66					
90-94	57					
95-99	27					
100-109	13					

a) Calculate the:-

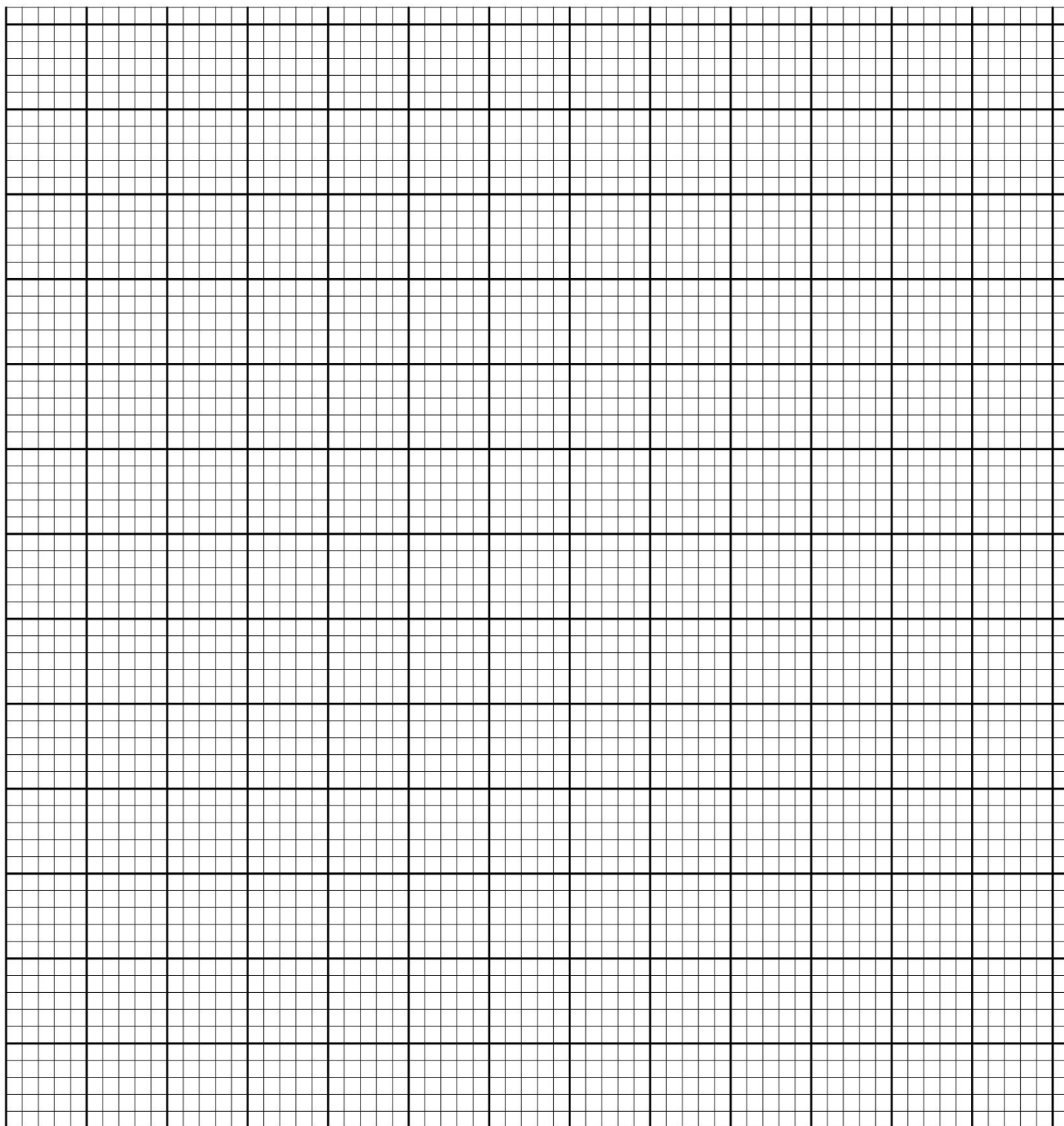
i) Quartile deviation

3marks

b) Standard deviation using a working mean of 102

3marks

- c) Draw an ogive curve representing the above information and use it to determine the percentage number of seedling whose height ranges between $85 \leq H \leq 93$. **(4marks)**

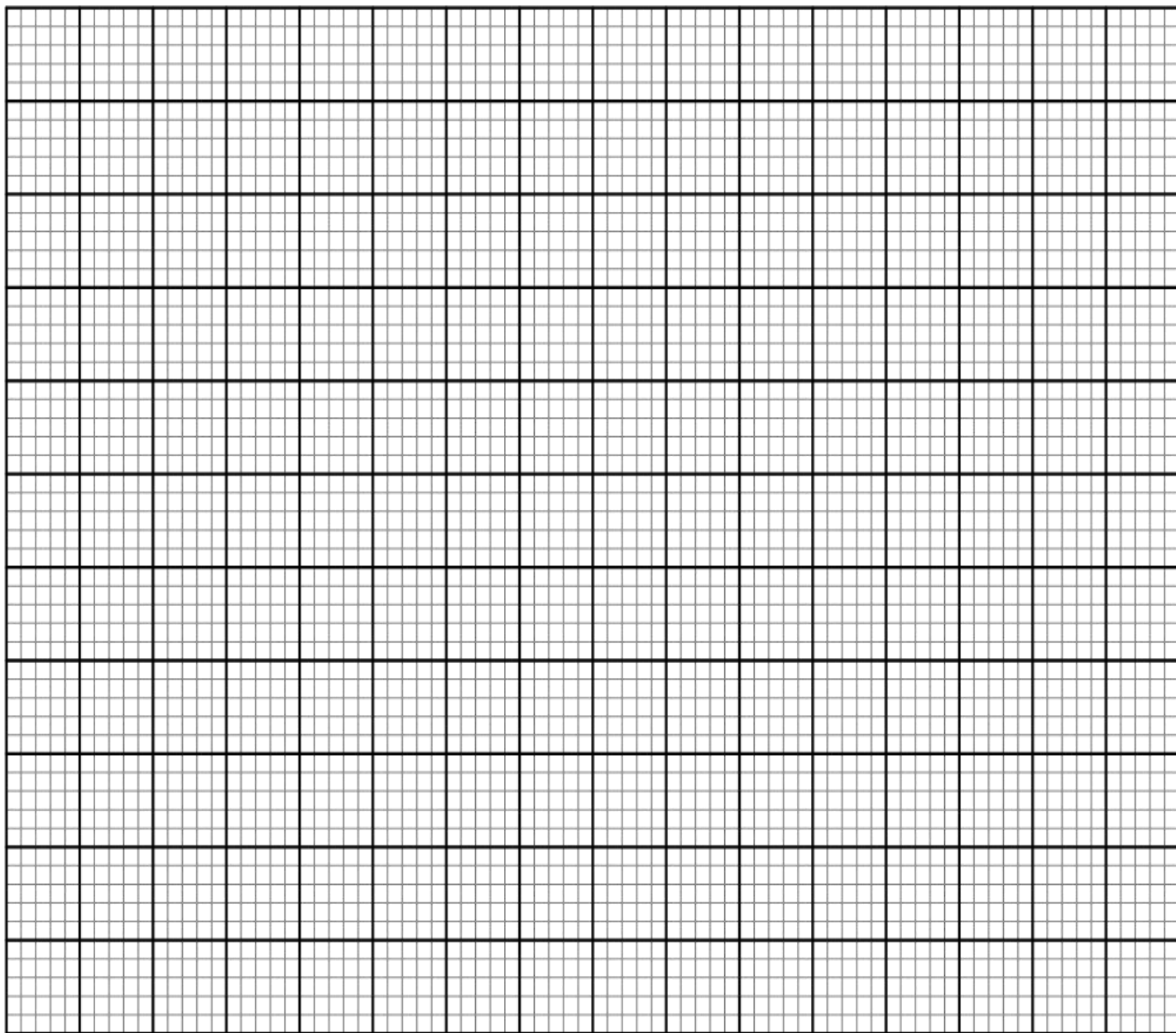


19. In order to ensure optimal health a lab technician needs to feed the rabbits on a daily diet containing a minimum of 24grams of fat, 36grams of carbohydrates and 4grams of protein. Rather than order rabbit food that is custom blended it is cheaper to order food X and food Y and blend them for an optimal use. One packet of food X contains 6grams of fat ,12 grams of carbohydrates ,2 grams of proteins and costs Sh 50. While one packet of food Y contains 12grams of fat,12 grams of carbohydrates,1 gram of proteins and it costs Sh 60

a). Form all the inequalities to represent the information above. (4 marks).

b). Graph all the inequalities
(4marks)

c). Determine the number of packets of type X and Y feed that should be used for optimal health at minimum cost.
(2marks)



20. Mr Kibet's PAYE per month is ksh 29,522. The ratio of his basic salary to taxable allowances is 13: 9. He is housed by his employer and pays a nominal rent of ksh 9000. His taxable pay is his gross pay plus 15% of his gross pay less the nominal rent. He is entitled to a personal relief of ksh 1056 per month. The following tax rates are used in the tax computation.

Income per annum (k£ per annum)	Rates (Ksh per K£)
1-5808	2
5809-11280	3
11281-16756	4
16753-22224	5
22225 and over	6

Calculate:

a) His basic salary per month

7marks.

b) The percentage change in his PAYE if his taxable pay increases by 10%

3marks

21. A group of students wish to build up a pile of toy bricks so as to have 2 bricks in the top row, 4 bricks in the second row, and 6 bricks in the third row and so on. If they have 3000 bricks

a) How many rows can they complete and how many bricks are they left with? **(4marks)**

b) Calculate the least number of bricks they should add to have all bricks used with no remainder?

(2marks)

c) Counting from the top, the 3rd row, 9th row, 27th row and so on form a geometric progression. If rows forming this progression are to be summed up, calculate the total number of bricks forming this arrangement after the number of bricks in (b) above have been included. **(4marks)**

22. P and Q are two points on latitude $60^{\circ}S$. Their longitudes are $30^{\circ}E$ and $150^{\circ}W$ respectively. Find to one decimal place : (Take the radius of the earth = $6370km$ and $\pi = \frac{22}{7}$)
- a) The distance in km between P and Q along the parallel of latitudes. (2marks)

- b) The shortest distance along the earth's surface between P and Q. (3marks)

- c) A weather forecaster reports that the center of a cyclone at ($30^{\circ}S$ $120^{\circ}W$) is moving due south at 24 knots. How long will it take to reach a point ($45^{\circ}S$, $120^{\circ}W$). (2marks)

- d) A plane leaves P at 1400hrs to Q at a speed of 900 knots along the shortest route. Determine the time at Q when the plane arrived. (3marks)

23. A curve is represented by the following function; $y = x^3 - 4x^2 + 5x - 2$. Use the curve to answer the following questions;

a). Find $\frac{dy}{dx}$ (1 mark)

b). (i). Determine the values of the stationery points of the curve. (4marks)

(ii). Determine the nature of each stationery point on the curve (2 marks)

c). In the space provided, sketch the curve of $y = x^3 - 4x^2 + 5x - 2$ (3 marks)

24. Three quantities A, B and C are such that A varies directly as the square root of B and inversely as the square of C .

a). Given that $A = 4$ when $B = 64$ and $c = 5$, find

i). The law connecting A, B and C **(4 marks)**

ii). A when $B = 16$ and $C = 10$ **(2 marks)**

b). If B is increased by 44% and C decreased by 20%, find the percentage change in A . **(4 marks)**

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